

LAB MANUAL

Experiment: Project Management Tool using Trello

Aim:

To understand the basics of project management and to plan, track, and manage a software project using a project management tool (Trello).

Objectives:

1. To understand project management concepts
2. To use a project management tool
3. To create and manage project tasks
4. To track project progress using workflow boards

Theory:

Project management involves planning, organizing, and managing resources to successfully complete project goals. Project management tools help teams collaborate, assign tasks, monitor progress, and meet deadlines efficiently.

Trello is a Kanban-based project management tool that uses boards, lists, and cards to represent projects and tasks visually.

Software Requirements:

- Web Browser (Chrome/Edge/Firefox)
- Internet Connection
- Trello Account

Procedure:

- Step 1: Open a web browser and go to trello.com
- Step 2: Login or sign up using your email ID
- Step 3: Click on Create → Create Board
- Step 4: Name the board as "Student Management System"
- Step 5: Create lists: To Do, In Progress, Testing, Done
- Step 6: Add task cards under To Do list
- Step 7: Assign due dates and members to cards
- Step 8: Move cards across lists as task progress changes

Part A: Agile Setup

1. Create a new board named **“Agile Project – Student Management System”**
2. Create lists:
 - Product Backlog

- Sprint Backlog
 - In Progress
 - Testing
 - Done
3. Add Product Backlog cards:
- User Login Module
 - Student Registration
 - Course Management
 - Attendance Tracking
 - Result Generation
4. Add user story description:
As a user, I want ____ so that ____
5. Move selected cards to **Sprint Backlog**
6. Add checklist to Sprint cards:
- Design
 - Coding
 - Unit Testing
 - Integration
7. Assign sprint due date, members, and sprint label

Part B: Automation using Butler

8. Click **Automation** → **Rules**
9. Create rule:
- When all checklist items are completed → Move card to **Done**
10. Create rule:
- When due date is marked complete → Move card to **Done**
11. Create rule:
- When label “Bug” is added → Move card to **Testing**

Part C: Monitoring

12. Enable **Calendar Power-Up**
13. Review completed sprint tasks
14. Conduct Sprint Retrospective

Expected Output

An Agile Scrum board with automated task movement and sprint visualization.

Expected Output:

A Trello board displaying tasks categorized under different workflow stages with progress tracking.

Result:

Thus, the project management tool Trello was successfully used to plan and manage a software project.

Conclusion:

1. What is project management?
2. What is Trello?
3. What is Kanban?
4. Advantages of project management tools
5. Difference between task and milestone

1. What is Project Management?

At its simplest, **Project Management** is the practice of leading a team to achieve specific goals and meet success criteria within a specified timeframe. It involves planning, executing, monitoring, and closing projects.

Unlike "operations" (which are ongoing and repetitive), a **project** has a defined beginning and end. Project management uses specific processes and tools to ensure that the "Triple Constraint"—**Scope, Time, and Cost**—is balanced effectively.

2. What is Trello?

Trello is a popular, visual collaboration tool that organizes your projects into boards. It is essentially a digital version of a whiteboard covered with sticky notes.

- **Boards:** The entire project workspace.
- **Lists:** Columns that represent different stages of a process (e.g., "To Do," "Doing," "Done").
- **Cards:** Individual items or tasks that live within lists and can be moved from one stage to the next.

3. What is Kanban?

Kanban is a visual workflow management method originated by Toyota to improve manufacturing efficiency. In the world of software and general project management, it focuses on "Just-in-Time" delivery.

The core of Kanban is the **Kanban Board**, which helps teams visualize their work, limit "Work in Progress" (WIP), and maximize efficiency. It differs from other methods (like Scrum) because it is continuous and doesn't usually use fixed-length "sprints."

4. Advantages of Project Management Tools

Using tools like Trello, Jira, or Asana offers several benefits over manual tracking:

- **Centralized Communication:** Keeps all files, comments, and updates in one place rather than buried in emails.
- **Improved Transparency:** Everyone on the team knows exactly who is doing what and when it's due.
- **Task Prioritization:** Helps identify high-impact tasks so the team doesn't waste time on low-priority work.
- **Remote Collaboration:** Essential for distributed teams to stay synced in real-time.
- **Resource Management:** Prevents team burnout by ensuring no single person is overloaded with too many tasks.

• Feature	• Task	• Milestone
• Definition	• A specific action or piece of work to be completed.	• A significant point or event in the project timeline.
• Duration	• Has a start and end date (takes time to do).	• Zero duration (it is a point in time).
• Example	• "Write the first chapter of the research paper."	• "Research paper draft completed."
• Purpose	• To track the actual work being done.	• To track progress against the overall schedule.

Output:











